

Chapter 3. **GROUPS**

3.3 Cyclic Groups

3.3.2 The Fundamental Theorem of Finite Cyclic Groups



In this video, we will prove a very useful result needed to draw the lattice diagram of the subgroups of a finite cyclic group

Suppose that $G = \langle g \rangle$ is a cyclic group of order n , and k and m are both positive divisors of n . Then we have that

$$\langle g^m \rangle \subseteq \langle g^k \rangle \Leftrightarrow k|m$$



$\Rightarrow$ | Suppose first that $\langle g^m \rangle \subseteq \langle g^k \rangle$, and let us show that $k|m$.

In particular, $g^m \in \langle g^k \rangle$. So,

$$g^m = (g^k)^t, \text{ for some integer } t.$$

But then $g^m = g^{kt}$ implies that

$$m \equiv kt \pmod{n}.$$

$$m \equiv kt \pmod{n} \Rightarrow n \mid (m - kt)$$

$$\Rightarrow m - kt = nu \text{ for some integer } u.$$

On the other hand from $k \mid n$, we have that $n = kv$ for some integer v .

We then have:

$$m - kt = nu = n(kv) \Rightarrow$$

$$\Rightarrow m = k(t + nv) \Rightarrow \underline{\underline{k \mid m}}$$

$\Leftarrow$ Suppose now that $k \mid m$ and let us show $\langle g^m \rangle \subseteq \langle g^k \rangle$. To show this, it is enough to prove $\underline{\underline{g^m \in \langle g^k \rangle}}$.

From $k \mid m$, we have that

$$m = ku$$

for some integer u . But then

$$g^m = g^{ku} = (g^k)^u \in \langle g^k \rangle.$$

